AUTOSAR HLD Document Analysis Assistant

Synopsis · Capstone Project, Case Study 1

Student	Kiran Soorya R S
Register number	126003133
Programme	B.Tech Computer Science and Engineering
University	SASTRA Deemed University

1. Problem statement and intended users

AUTOSAR High-Level Design (HLD) documents describe an ECU's software architecture: its software components (SW-Cs), port interfaces, ports, signals, dependencies and functional flows. They are long PDFs made mostly of tables, so answering a question such as "*which component consumes this interface?*" means cross-reading several tables by hand. Inconsistencies, such as an interface that no component provides or a dependency on an undefined component, are easy to miss. General-purpose chatbots are not a safe substitute: they answer from training data, cite no sources and can invent component names.

The intended users are system and AUTOSAR architects, software developers, integration and test engineers, and engineering managers who review or build on these designs.

2. Objectives

- Reduce the manual effort of reading and reviewing HLD documents.
- Build a structured, searchable inventory of the architecture.
- Answer natural-language questions only from the documents, with page and section citations, and refuse when the documents do not contain the answer.
- Detect inconsistencies and compare document revisions.
- Keep an engineer in control: every extracted element is reviewed, and every decision is logged.

3. Approach

1. **Ingestion.** A PDF is read page by page, section headings are detected, and table-of-contents pages are skipped. The text is split into 800-character passages that overlap by 150 characters.
2. **Local indexing.** Each passage, together with its section heading, is embedded locally by the all-MiniLM-L6-v2 model and stored in a FAISS index, so documents never leave the machine for indexing.
3. **Extraction.** Rule-based parsers read the document's tables into a SQLite inventory of SW-Cs, interfaces, ports, signals, dependencies and flows. A language model then refines the result, filling gaps, but it cannot add to an element type the document defines in a table.
4. **Grounded question answering.** A question retrieves the five closest passages. If none is close enough (squared L2 distance above 1.1), the question is refused without calling the model. Otherwise the model answers at temperature 0 under strict rules: use only the passages, cite document, page and section, and give a fixed refusal sentence if the answer is missing.
5. **Analysis.** Consistency checks (missing providers, undefined components, unused interfaces, dependencies not backed by ports) and revision comparison run on the inventory.
6. **Human review.** An engineer approves, rejects or corrects each element with a mandatory rationale. Decisions are stored in an append-only audit log and included in every export.

4. Technology

Python · Streamlit (user interface) · FastAPI (REST service) · pypdf · sentence-transformers (all-MiniLM-L6-v2) · FAISS · Mistral AI (ministral-8b-latest) · SQLite · pytest

5. Input data

A 24-page synthetic HLD of a Vehicle Brake Control System (6 SW-Cs, 10 interfaces, 29 ports, 11 signals, 19 dependencies, 8 functional flows), two revisions of a 2-page Electronic Brake Control HLD for comparison and consistency testing, and three public AUTOSAR specifications as background knowledge.

6. Key results

- **Extraction:** precision and recall of 1.00 for every element type on both HLDs; all counts match the document's own architecture summary table.
- **Grounding:** all 11 answerable test questions passed the evidence check, and 5 of 6 unanswerable questions were blocked before reaching the model.
- **End-to-end answers:** 10 of 11 answerable questions were answered correctly, all with inline citations, and all 6 unanswerable questions were refused. Answers took 1 to 2.5 seconds.
- **Analysis:** no false findings on the consistent document; every defect planted in the second revision was detected.
- **Quality:** 84 automated tests pass.

7. Conclusion and future work

The assistant turns long, table-heavy HLD documents into a searchable, checkable and reviewable architecture model, and answers questions only when the documents support the answer. Future work: OCR for scanned PDFs, an entailment check between answers and passages, role-based access control, and re-calibrating the evidence threshold on real project documents.